

Counting Permutations

Time limit: 1.00 s

Memory limit: 512 MB

A permutation of integers $1, 2, \dots, n$ is called *beautiful* if there are no adjacent elements whose difference is 1.

Given n , your task is to count the number of beautiful permutations.

Input

The only input line contains an integer n .

Output

Print the number of beautiful permutations of $1, 2, \dots, n$ modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 1000$

Example

Input	Output
5	14